

Module 4: Strategic Equilibrium applications & Dynamic games

ECO364-THINKING STRATEGICALLY

Part II: Dynamic games

THE EXTENSIVE FORM, PERFECT EQUILIBRIUM, FIRM-ENTRY DECISIONS,
SEQUENTIAL VOTING, CRITICISM OF THE PERFECT EQUILIBRIUM CONCEPT

Modeling dynamic games

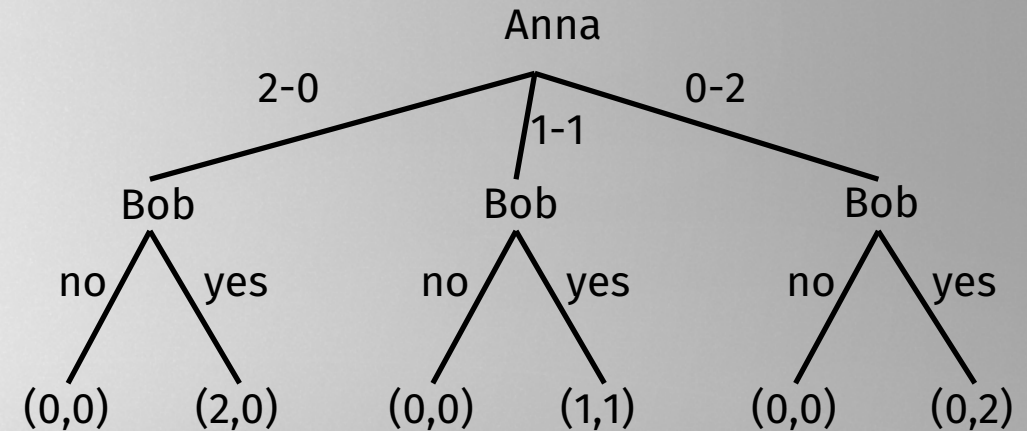
THE EXTENSIVE FORM (BASED ON A LECTURE BY PROF. DANA NAU)

Dynamic games

- ▶ In the previous chapters we analyzed mostly simultaneous strategic interactions where players choose their actions simultaneously.
- ▶ We next study sequential interactions where players make alternating moves.

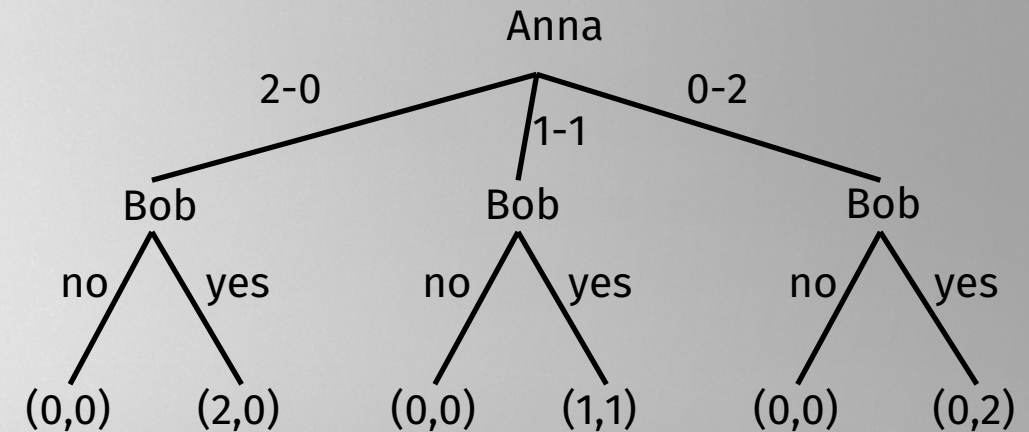
A cookie sharing game

- ▶ Suppose we have two logical children, Anna and Bob, who like cookies a lot.
- ▶ Their teacher offers them two cookies, but only if they agree how to share them.
- ▶ Anna can choose one of the following options:
 - ▶ Anna gets 2 cookies, Bob gets 0 cookies
 - ▶ They each get 1 cookie
 - ▶ Anna gets 0 cookies, Bob gets 2 cookies
- ▶ Bob chooses to accept or reject the split:
 - ▶ Accept => they each get their cookies(s)
 - ▶ Otherwise, neither gets any



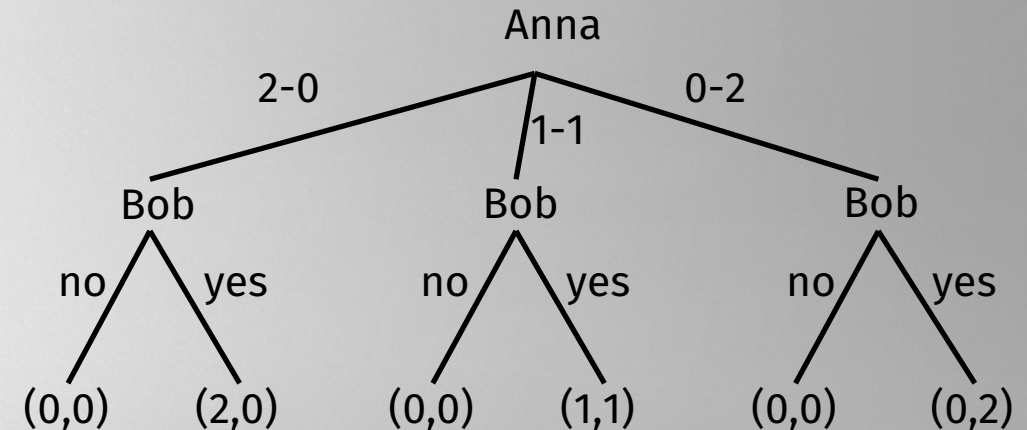
The extensive form of a game

- ▶ The cookie sharing game is a game in extensive form
 - ▶ A game representation that makes the temporal structure explicit
 - ▶ Doesn't assume agents act simultaneously
- ▶ Extensive form can be converted to normal form, we will see in a bit how.
 - ▶ So previous results about equilibrium carry over.
 - ▶ However, there are additional results that depend on depend on the sequential or dynamic nature of the game.
- ▶ Extensive form is a game tree:
 - ▶ Nonterminal node = place where an agent chooses an action
 - ▶ Terminal node = a final outcome
 - ▶ At each terminal node, each player has a payoff.



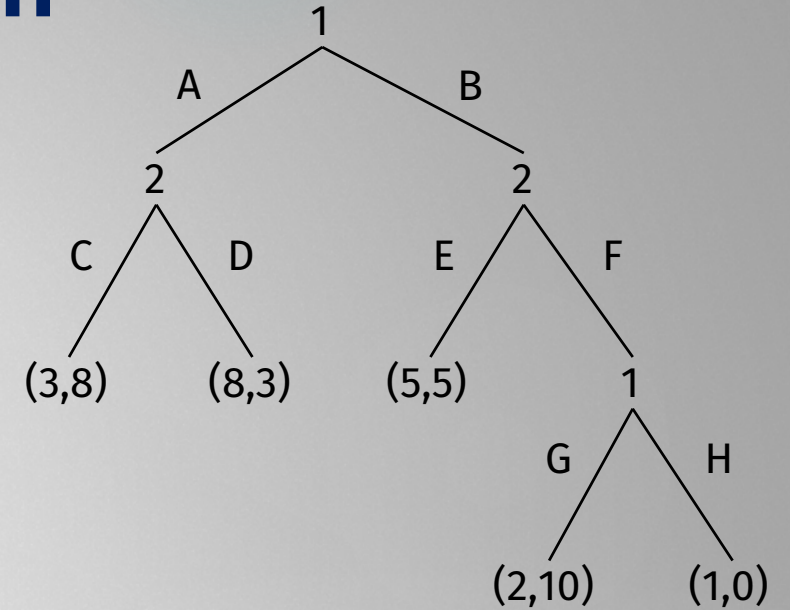
The extensive form: pure strategies

- ▶ A pure strategy for a player in an extensive form game:
 - ▶ Specifies which action to take at every node where it is the player's choice
 - ▶ Remember, a **strategy** is a **complete and contingent plan of action**.
 - ▶ This means that a strategy should specify how a player chooses an action at each of his or her nodes (complete) and at the same time respecting the sequential structure of the game, i.e. respecting what other players that moved before did (contingent).
- ▶ Cookie sharing game:
 - ▶ Anna has 3 pure strategies:
 - ▶ $S_A = \{2-0, 1-1, 0-2\}$
 - ▶ Bob has 8 pure strategies:
 - ▶ $S_B = \{(yes, yes, yes), (yes, yes, no), (yes, no, yes), (yes, no, no), (no, yes, yes), (no, yes, no), (no, no, yes), (no, no, no)\}$
 - ▶ A strategy for Bob specifies what to do at each possible node in which he chooses and also respects the fact that Anna moved first.



Extensive form vs. normal form

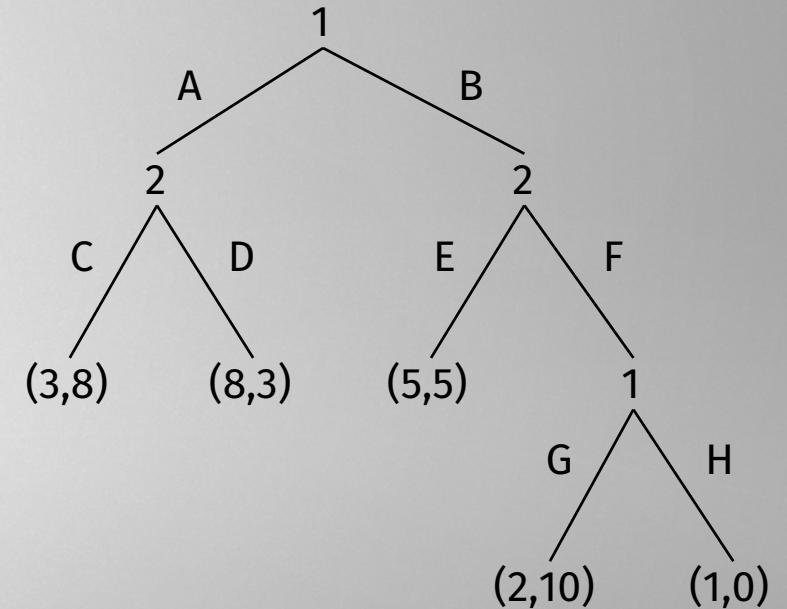
- ▶ Every game tree corresponds to an equivalent normal-form game
- ▶ The first step is to get all of the agents' pure strategies
- ▶ An agent's complete strategy must specify an action at every node where it's the agent's move
- ▶ Example: the game tree shown here
 - ▶ Agent 1 has four pure strategies:
 - ▶ $S_1 = \{(A, G), (A, H), (B, G), (B, H)\}$
 - ▶ Must include (A, G) and (A, H), even though action A makes the G-versus-H choice moot. Remember, a strategy is a complete and contingent plan of action.
 - ▶ Agent 2 also has four pure strategies:
 - ▶ $S_2 = \{(C, E), (C, F), (D, E), (D, F)\}$



Extensive form vs. normal form

- Once we have all of the pure strategies, we can rewrite the game in normal form:

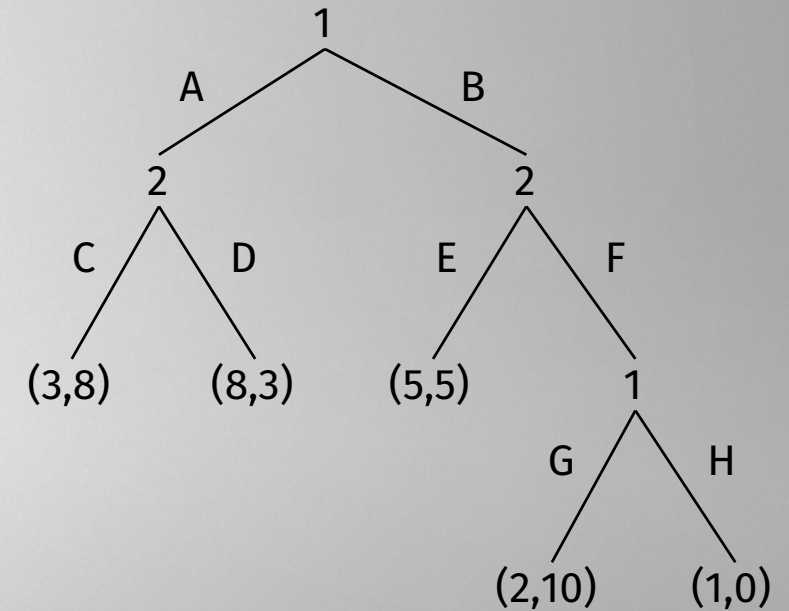
	(C,E)	(C,F)	(D,E)	(D,F)
(A,G)	3,8	3,8	8,3	8,3
(A,H)	3,8	3,8	8,3	8,3
(B,G)	5,5	2,10	5,5	2,10
(B,H)	5,5	1,0	5,5	1,0



Nash equilibrium

- ▶ With the normal form, we can find the Nash Equilibria:
 - ▶ There are three pure strategy equilibria in this game.
 - ▶ The profiles $\{(B,H), (C,E)\}$, $\{(A,G),(C,F)\}$ and $\{(A,H),(C,F)\}$ are equilibria.

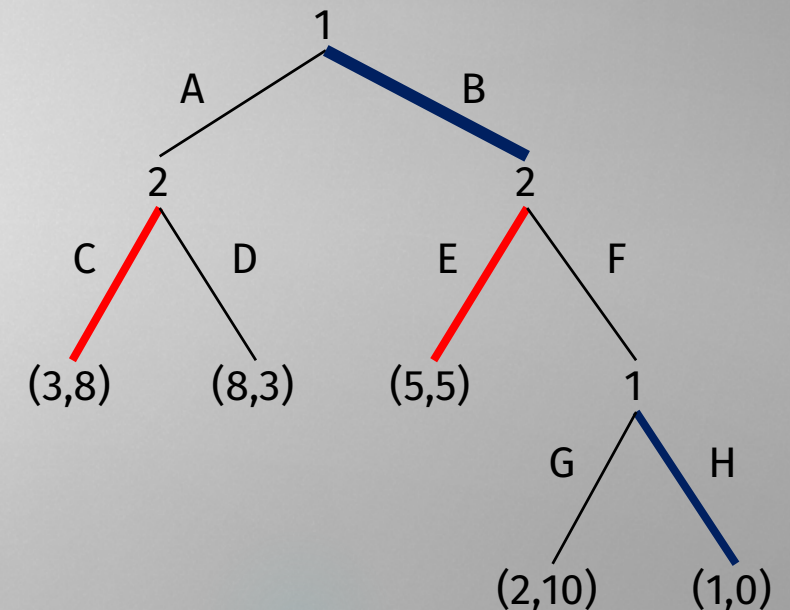
	(C,E)	(C,F)	(D,E)	(D,F)
(A,G)	3,8 	 3,8 	 8,3	 8,3
(A,H)	3,8 	 3,8 	 8,3	 8,3
(B,G)	 5,5	2,10 	5,5	2,10 
(B,H)	 5,5 	1,0	5,5 	1,0



Nash equilibrium

- ▶ So far, we have used what we learned in the previous modules.
- ▶ Recall that the Nash Equilibria in this game are:
 - ▶ $\{(B,H), (C,E)\}$
 - ▶ $\{(A,G),(C,F)\}$
 - ▶ $\{(A,H),(C,F)\}$
- ▶ Let's depict the equilibrium $\{(B,H), (C,E)\}$ in the extensive form.
 - ▶ It says that 1 plays B in her first node and H in her second node.
 - ▶ It also says that 2 plays C in his first node and E in his second.
- ▶ Do you see something problematic with this equilibrium?

	(C,E)	(C,F)	(D,E)	(D,F)
(A,G)	3,8	3,8	8,3	8,3
(A,H)	3,8	3,8	8,3	8,3
(B,G)	5,5	2,10	5,5	2,10
(B,H)	5,5	1,0	5,5	1,0

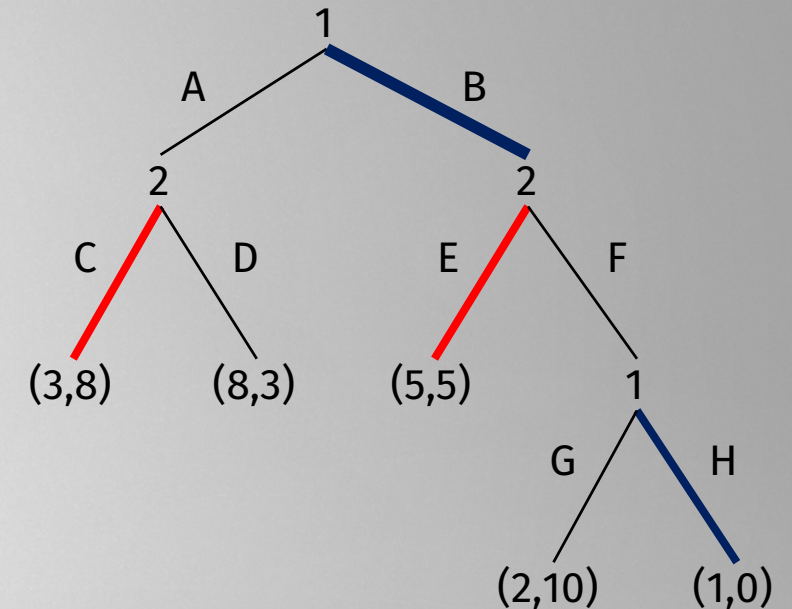


A refinement of Nash Equilibrium

SUBGAMES, SUBGAME PERFECT EQUILIBRIUM AND BACKWARDS INDUCTION

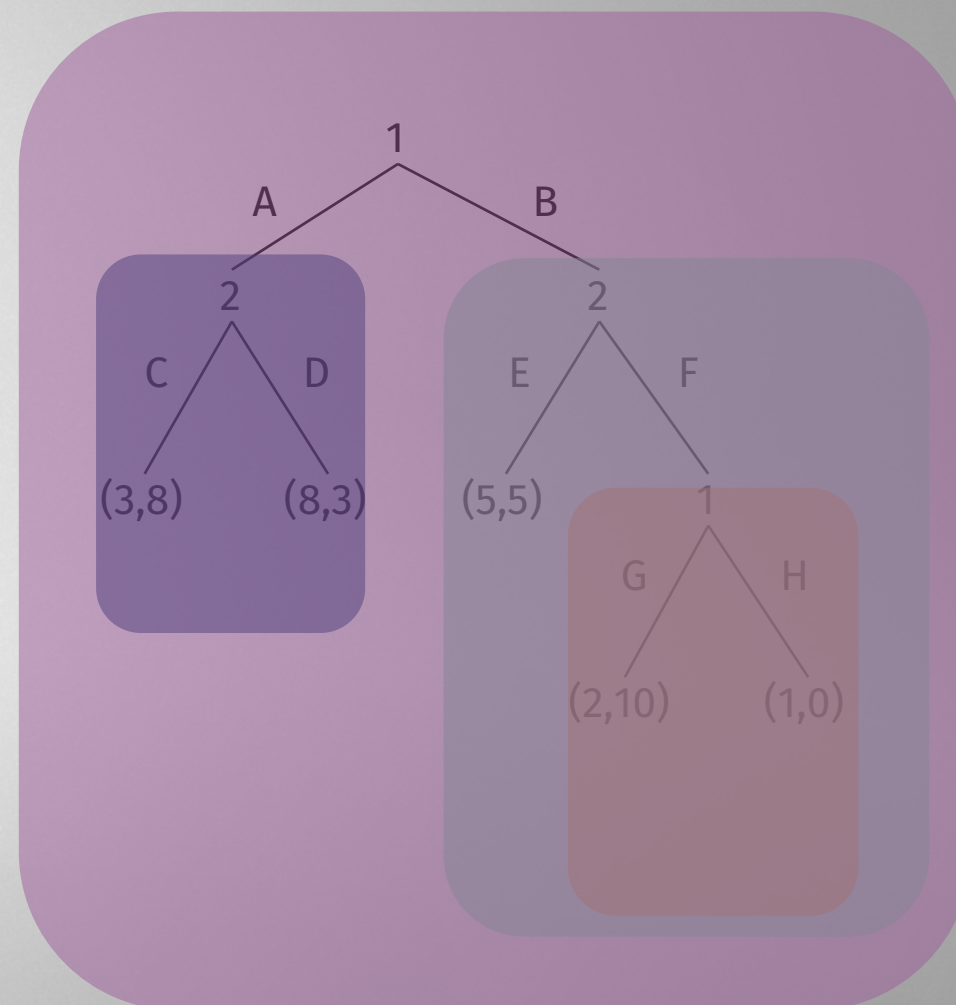
Nash equilibrium: too weak for dynamic games

- ▶ If agent 1 used (B,G) instead of (B,H)
 - ▶ Then agent 2's best response would be (C,F), not (C,E)
- ▶ When agent 1 plays B
 - ▶ The only reason for agent 2 to choose E is if agent 1 has already committed to H rather than G.
- ▶ This behavior by agent 1 is a **threat**:
 - ▶ By committing to choose H, which is harmful to agent 2, agent 1 can make agent 2 avoid that part of the tree
 - ▶ Thus agent 1 gets a payoff of 5 instead of 2
- ▶ But is agent 1's threat credible?
 - ▶ If agent 2 plays F, would agent 1 **really** play H rather than G?
 - ▶ It would reduce agent 1's own utility: is not a **sequential best response**!
 - ▶ If the game has reached the second node of agent 1, is not rational for her to commit to the threat of playing H, since it is in her best interest to play G!



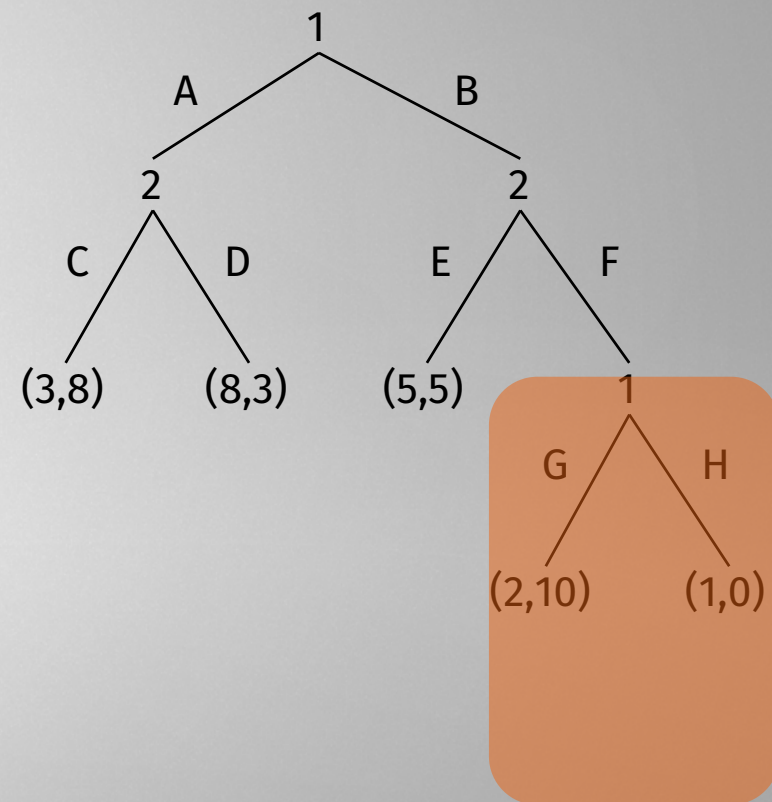
Subgame Perfect Equilibrium

- ▶ This equilibrium concept is a refinement of Nash Equilibrium.
- ▶ By refinement, I mean that we impose more restriction to the set of Nash Equilibria to pick the ones that satisfy this restriction.
- ▶ We call a **subgame** of an extensive form game G a node belonging to G *together with all its descendants*.
 - ▶ In the previous example the node corresponding to the second move of player 1 together with its descendants is a subgame
 - ▶ The right node where player 2 moves and all its descendants is another subgame.
 - ▶ The left node where player 2 moves and its descendants is another subgame.
 - ▶ The node where player 1 first moves together with all its descendants is also a subgame (the whole game satisfies the definition of a subgame).
 - ▶ These are all the subgames in this game.
- ▶ A subgame perfect equilibrium, or in short just perfect equilibrium, is a strategy profile such that for every subgame of an extensive form game G , the restriction of this profile to the subgame is also a Nash Equilibrium of the subgame.
- ▶ Intuitively, a perfect equilibrium, is an equilibrium profile that induces a Nash Equilibrium in every subgame of the original game.



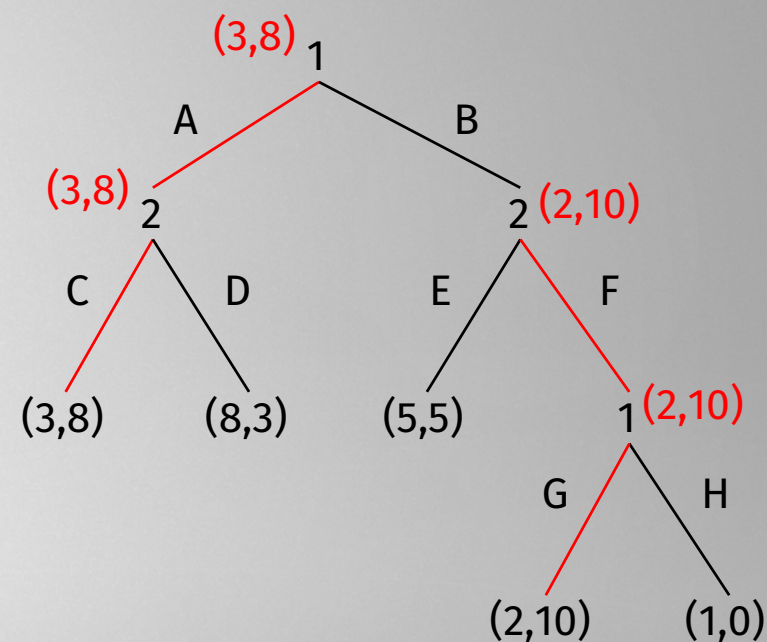
Subgame Perfect Equilibrium

- ▶ Recall that the Nash Equilibria in this game are:
 - ▶ $\{(B,H), (C,E)\}$
 - ▶ $\{(A,G),(C,F)\}$
 - ▶ $\{(A,H),(C,F)\}$
- ▶ Consider the subgame in orange.
 - ▶ For agent 1, G strictly dominates H
 - ▶ Thus H can't be part of a **Nash equilibrium** in this subgame.
 - ▶ This excludes $\{(A, H), (C, F)\}$ and $\{(B, H), (C, E)\}$: these Nash Equilibria involve **non-credible threats** (play H if the game ever reaches that node of player 1).
 - ▶ Just one subgame-perfect equilibrium: $\{(A, G), (C, F)\}$



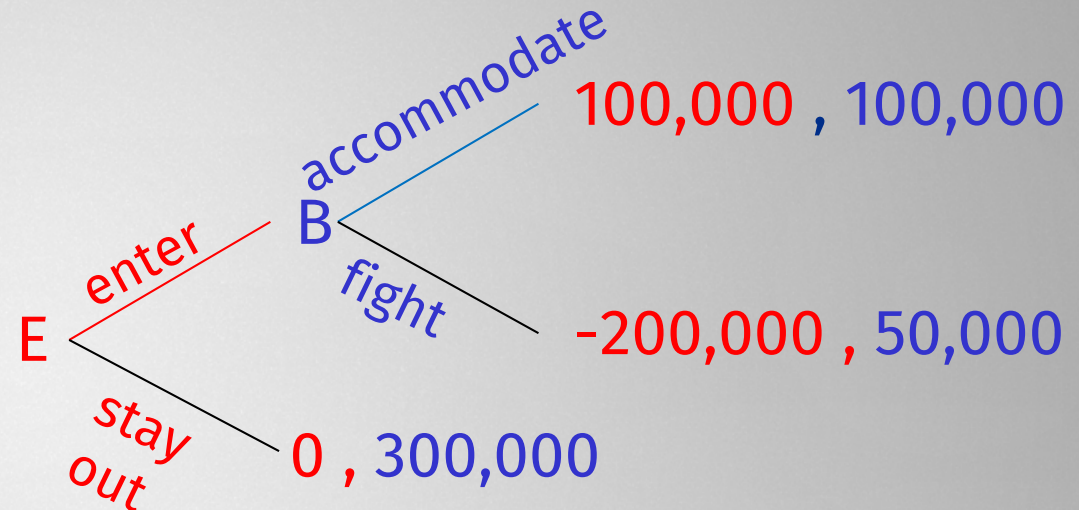
Finding SPE: Backwards Induction

- ▶ To find subgame-perfect equilibria, we can use backward induction
- ▶ Identify the equilibria in the bottom-most nodes
 - ▶ Assume they'll be played if the game ever reaches these nodes.
 - ▶ Delete the subgame and replace the payoffs that these equilibria generate in the new terminal nodes.
 - ▶ Go back to the first point and repeat.
- ▶ The backwards induction procedure identifies the subgame perfect equilibrium payoff (3,8) and the strategy profile that generates it: $\{(A, G), (C, F)\}$ which is represented by the red edges in the diagram.



Example: Entry deterrence by a firm

- ▶ A brand company **B** controls the market for air condition units. A new company **E** is about to enter this market. If **E** enters, **B** has two choices: accommodate **E** by accepting a lower market share or fight a price war.
- ▶ If **B** accommodates the entry, **E** will make a profit of \$100,000 but if **B** starts a price war, it will cause **E** to lose \$200,000. If **E** stays out its profit is zero. The outcome for **E** is described by the following tree:
- ▶ Suppose next that **B** as a monopolist, is able to make a profit of \$300,000. Sharing the market with **E** reduces its profit to \$100,000. Fighting a price war will leave **B** with a profit of \$50,000.
- ▶ Looking ahead and reasoning back we can see that if **E** enters then the best reply strategy of **B** is to accommodate **E** (since $100,000 > 50,000$). **E** takes this into account and chooses to enter to obtain 100,000.



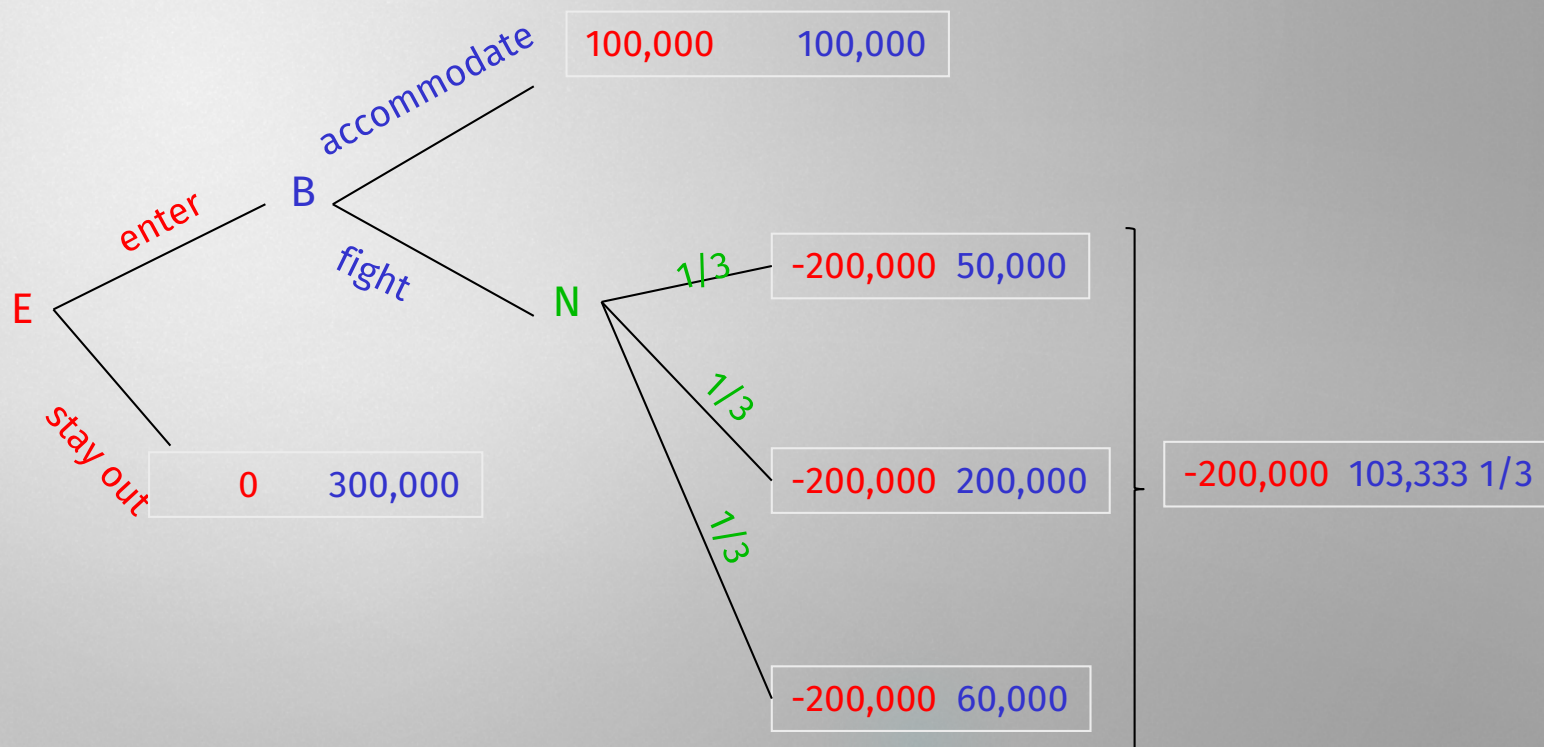
- ▶ Notice that this implies the SPNE profile {Enter, Accommodate}
- ▶ As an exercise, construct the normal form: you will see that there are two pure strategy NE {Enter, Accommodate} which is also the perfect equilibrium and another equilibrium which involves a non-credible threat by player B: {Stay Out, Fight}

Examples

FIRM ENTRY DETERRENCE, MORE GAMES & SEQUENTIAL VOTING

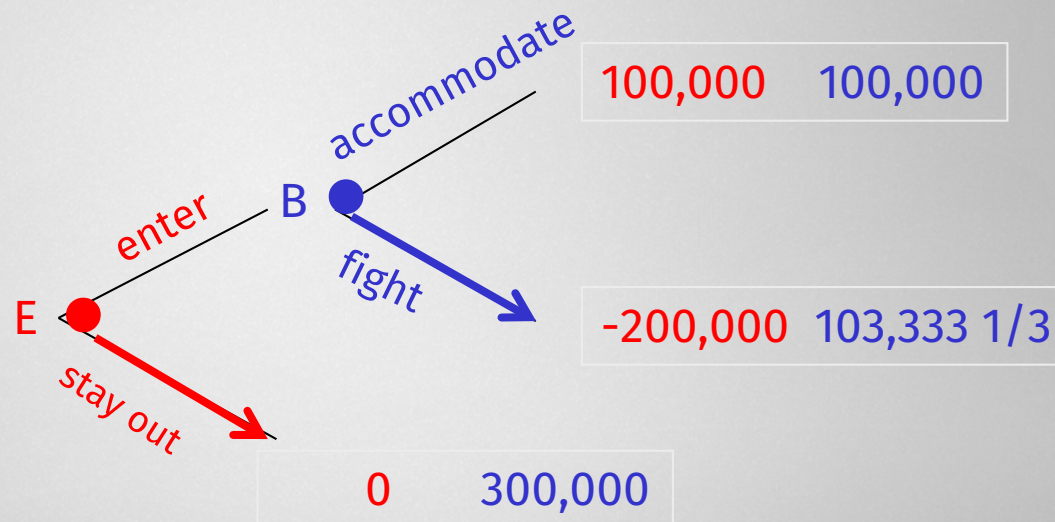
Example: Entry deterrence by a firm

- ▶ Suppose next that there is uncertainty about the profit of **B** if **E** enters. Suppose that **E** believes that in case **B** fights, **B** will obtain \$50,000 with probability $1/3$, \$200,000 with probability $1/3$ and with the remaining probability of $1/3$ **B** will obtain \$60,000. If **B** accommodates **E** then it will obtain \$100,000 with certainty.
- ▶ The game tree is:
- ▶ It involves a move by Nature (**N**).
- ▶ The expected payoff of this chance move by nature is depicted also.



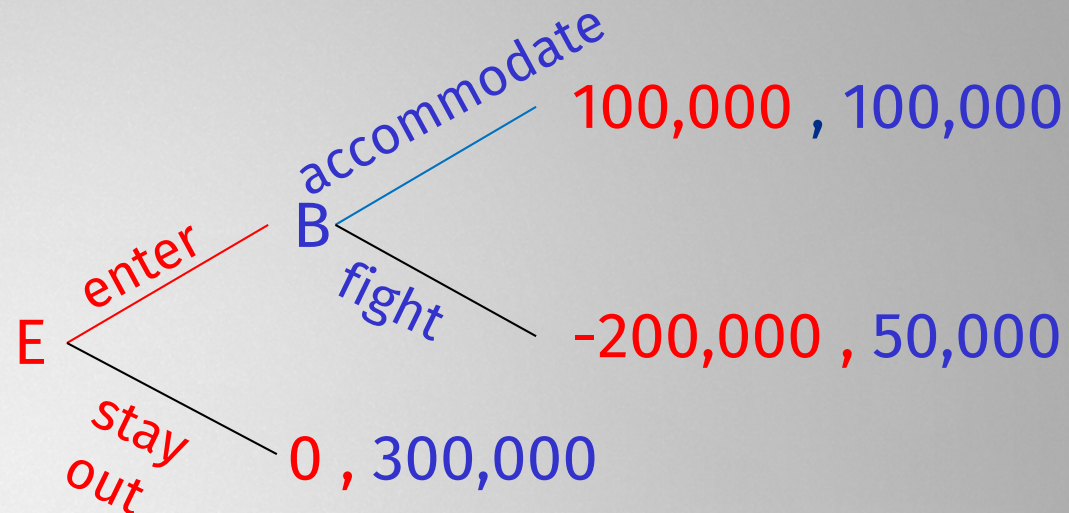
Example: Entry deterrence by a firm

- ▶ If **B** fights it will obtain an expected profit of $\frac{1}{3} \cdot 50,000 + \frac{1}{3} \cdot 200,000 + \frac{1}{3} \cdot 60,000 = 103,333 \frac{1}{3}$
- ▶ If **B** accommodates it obtains only \$100,000.
- ▶ The resulting tree is:
- ▶ Hence **B** will fight and **E** stays out
- ▶ Thus, the SPNE profile is:
 - ▶ {Stay Out, Fight}



Example: Entry deterrence by a firm

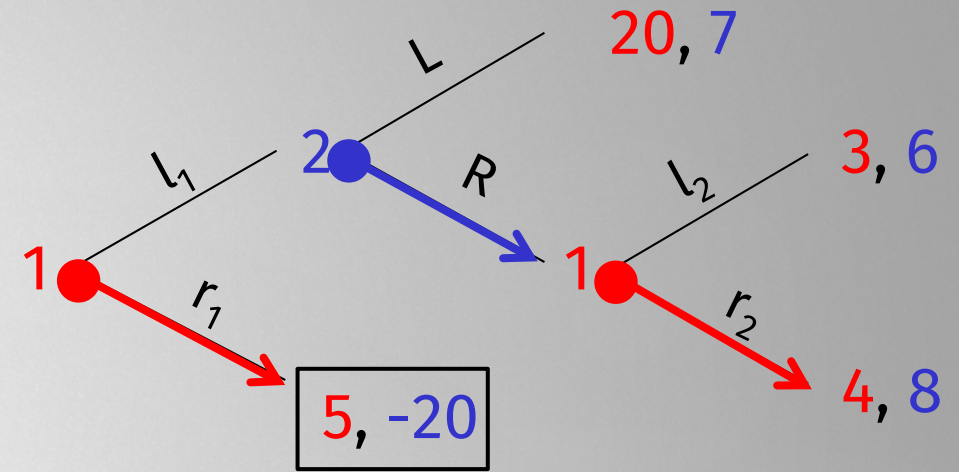
- ▶ Solving dynamic games (games in a tree form) by backward manner always yield a Nash equilibrium (a claim that need to be proven!). To see this let us examine the first example about entry deterrence:
- ▶ The strategic form of this game (payoffs in the thousands) is:
- ▶ There are here two Nash equilibrium points:
 - (1) (Enter, Accommodate) $\rightarrow$ (100, 100)
 - (2) (Out, Fight) $\rightarrow$ (0, 300)
- ▶ But only one of them, the first point, is obtained by backward induction.
- ▶ The other equilibrium is obtained when B threatens to fight any entry and E the potential entrant, takes the threat seriously and retreats.
- ▶ The threat of B is not credible. If E enters and B fights then E obtains 50,000. But if B accommodates E it obtains 100,000. Thus E should not believe the threat of B to fight and E should enter. Notice that the strategy "Fight" of B is indeed weakly dominated by its strategy "Accommodate".
- ▶ The Nash equilibrium which is obtained by backward induction manner is the Subgame Perfect Nash Equilibrium.



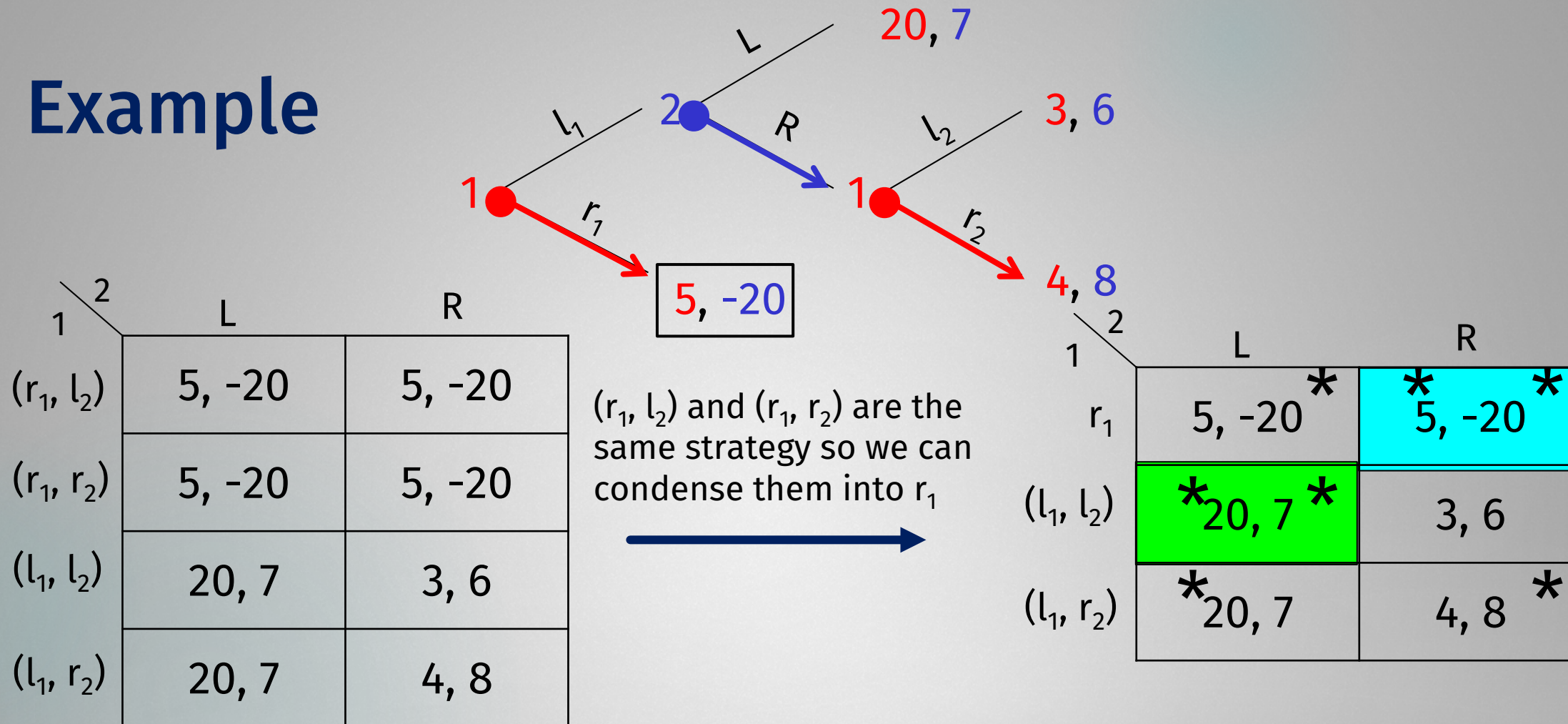
		B	
		Acc	Fight
E	Enter	* * 100, 100	-200, 50
	Out	0, 300 *	0, 300 *

Example

- ▶ Consider the following game in a tree form
- ▶ The perfect equilibrium is $((r_1, r_2), R) \rightarrow (5, -20)$
- ▶ Are there other equilibrium points in the game?
- ▶ Let us reduce this game to a game in strategic form. Note that 1 has four strategies:
 - ▶ (l_1, l_2)
 - ▶ (l_1, r_2)
 - ▶ (r_1, l_2)
 - ▶ (r_1, r_2)



Example



As we can see, we have two equilibrium points ((1) is the PE point):

(1) $(r_1, R) \rightarrow (5, -20)$

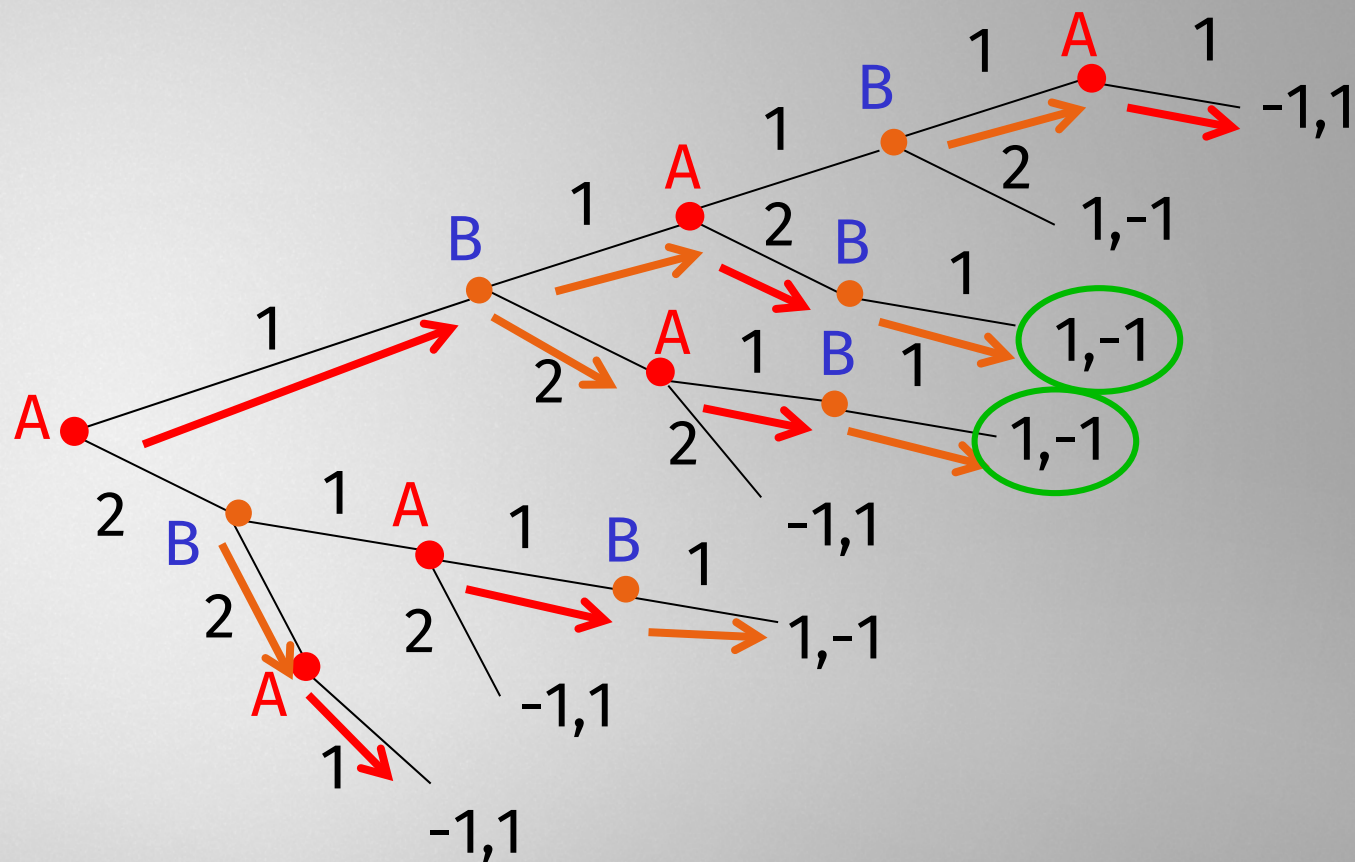
(2) $((l_1, l_2), L) \rightarrow (20, 7)$

Even though both players prefer the equilibrium (2), it is not sensible and will not be chosen. Notice that the strategy (l_1, l_2) of 1 is weakly dominated (by (l_1, r_2)) and therefore 1 will not choose it. Also observe that the equilibrium $((l_1, l_2), L)$ is based on a non credible threat of 1 to play l_2 if 2 plays R. But if 2 indeed plays R why should 1 play l_2 ? He obtains more if he plays r_2 .

The only sensible equilibrium outcome is (5, -20), which corresponds to the SPNE profile of $((r_1, r_2), R)$

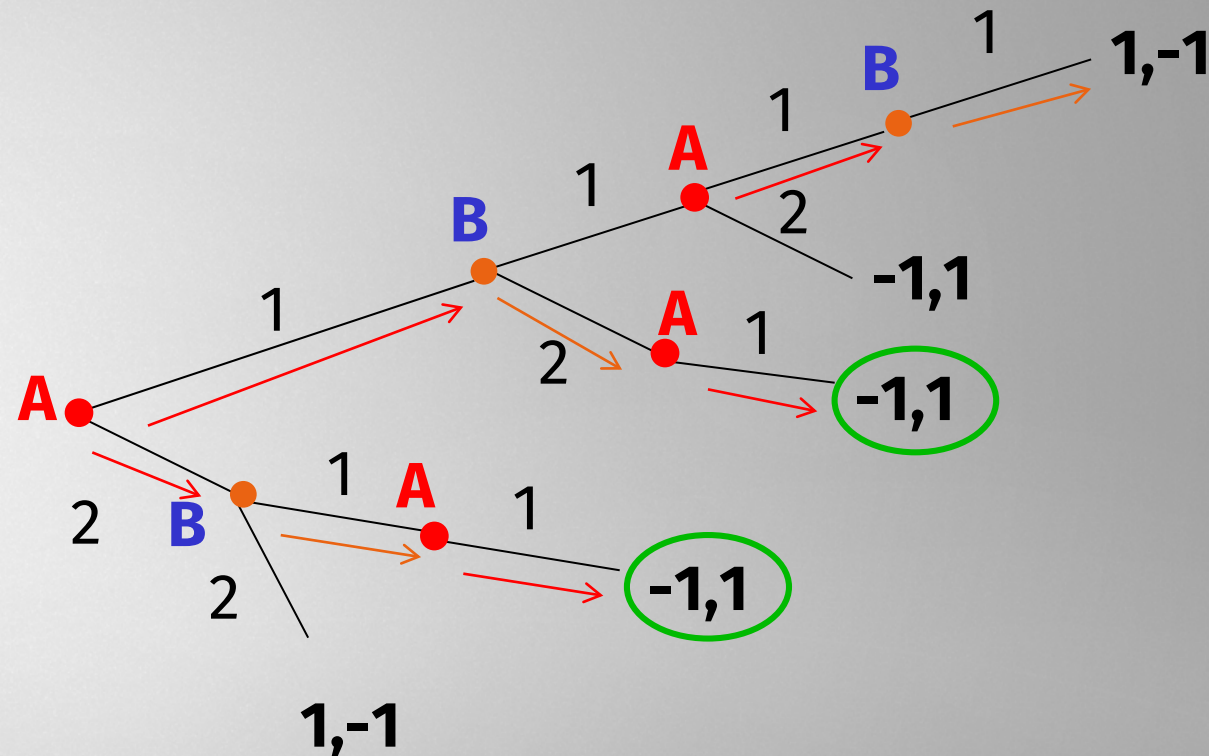
Example: Picking up bricks

- Five bricks have been stacked on the ground. The two players **A** and **B** take turns picking up either **one** or **two** bricks from the pile. The player who picks up the last brick **loses \$1**. The other player **wins \$1**.
- Solving backwardly, the winning strategy of the first mover **A** is to pick up in his first move one brick. **B** loses no matter what she does (if she picks up 2 bricks **A** will pick up 1 and, if she picks up 1 brick **A** will pick up 2 bricks).

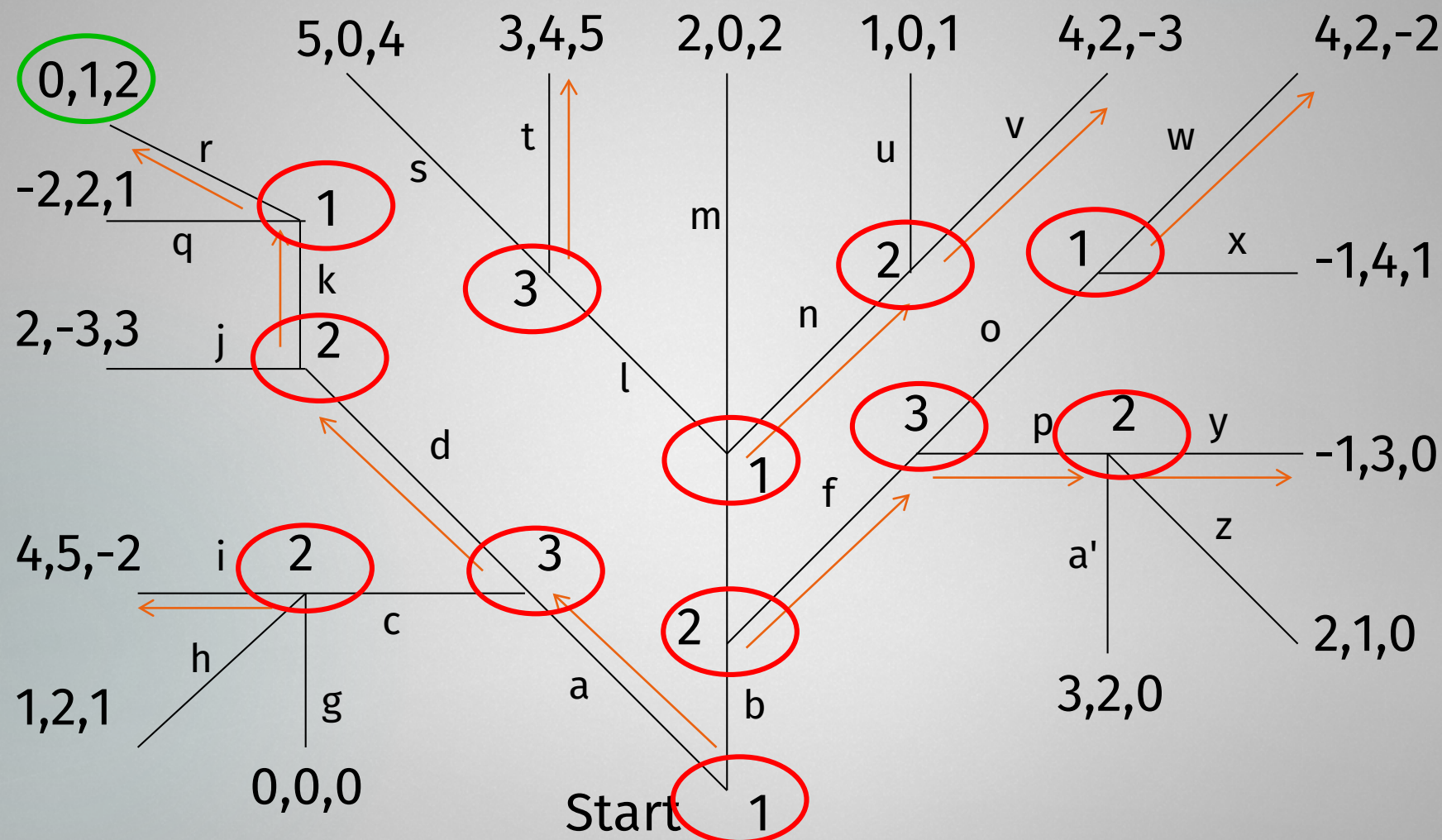


Example: Picking up bricks

- ▶ Can we conclude that the first mover has always the advantage?
- ▶ NO!
- ▶ Consider next the same example but now with four bricks only
- ▶ The follower B, has a winning strategy (if A picks up 1 brick B picks up 2 bricks and if A picks up 2 bricks, B picks up 1 brick). The second mover here has the advantage!
- ▶ Exercise: Can you analyze the general case with any number of bricks?



Example: 3-person game in tree form



- ▶ The perfect equilibrium outcome is $(0,1,2)$. The perfect equilibrium strategies are:
- ▶ Player 1: (a, r, n, w) , Player 2: (i, k, f, y, v) , Player 3: (d, t, p)

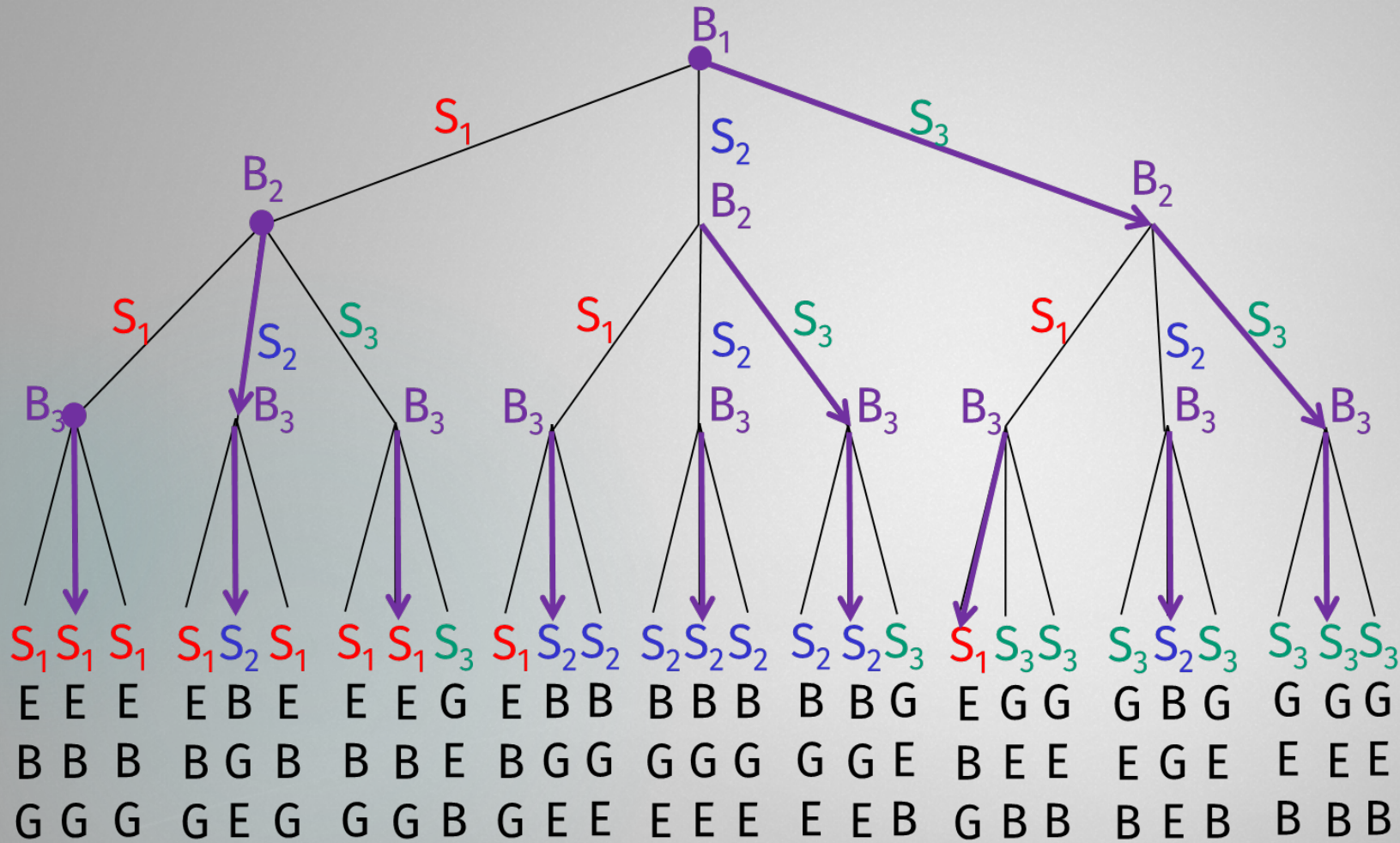
Sequential voting: Revisited

	B_1	B_2	B_3
S_1	E	B	G
S_2	B	G	E
S_3	G	E	B

E- Excellent
G- Good
B- Bad

- ▶ Suppose that the votes are open and taken in a sequence. First B_1 announces his choice, then B_2 and then B_3 .
- ▶ Claim: The winning strategy is S_3 .

Sequential Voting: Revisited



- ▶ The winning strategy is S_3 .
- ▶ The SPNE profile that delivers this winning strategy is denoted with purple edges.
- ▶ Notice how complicated is the strategy of B_3 : this board member needs to consider *every possible contingency* and make a plan for each one of them.

Example: 2-person game in tree form with moves from nature.

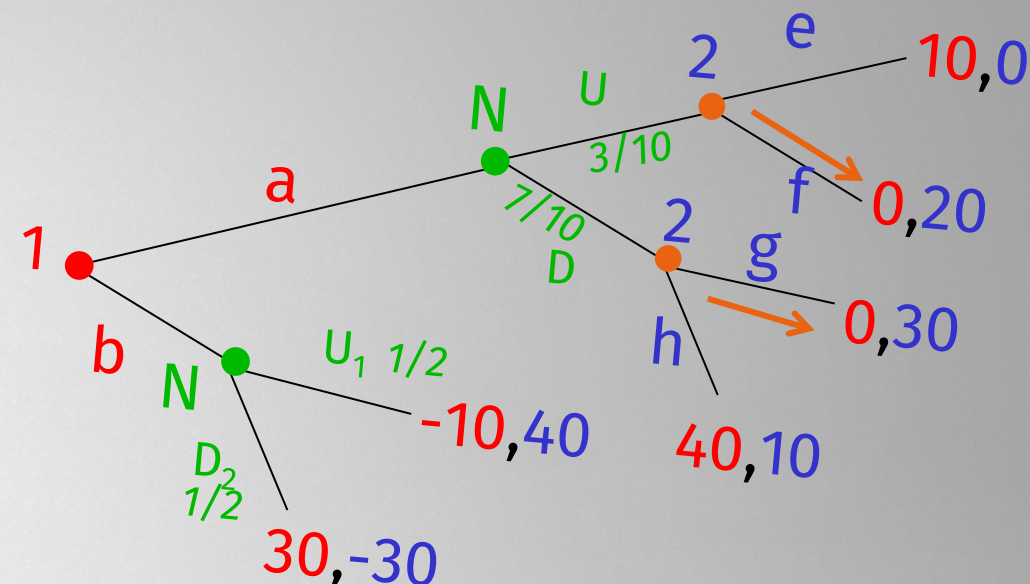
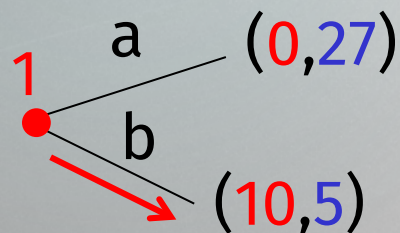
- ▶ Taking into account that Player 2 plays *f* and *g*, if 1 chooses *a* the expected payoffs of the players are:

$$\text{▶ } (3/10) \cdot (0, 20) + (7/10) \cdot (0, 30) = (0, 6) + (0, 21) = (0, 27)$$

- ▶ Also, if 1 chooses *b* the expected payoffs of the players are:

$$\text{▶ } (1/2) \cdot (-10, 40) + (1/2) \cdot (30, -30) = (-5, 20) + (15, -15) = (10, 5)$$

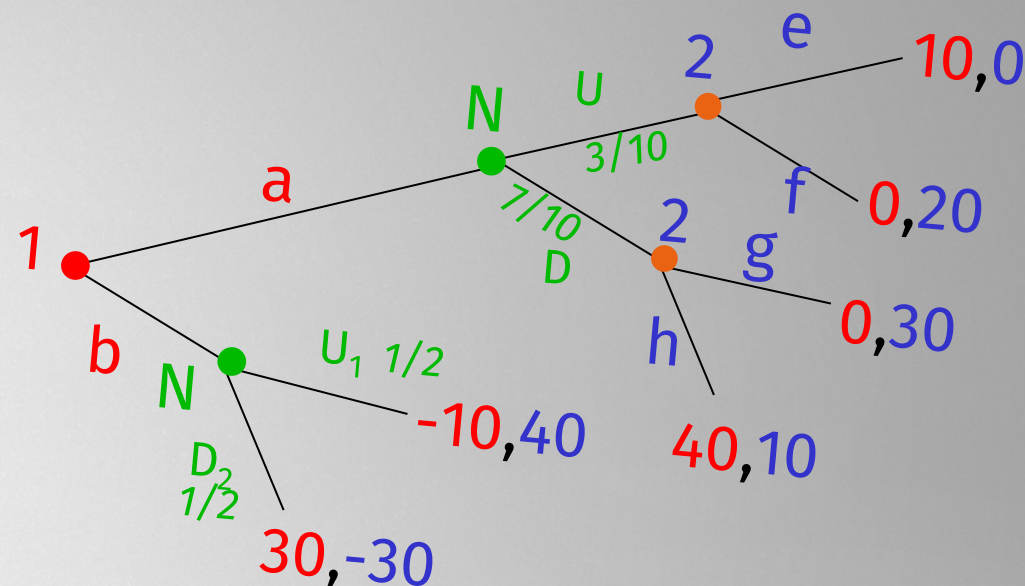
- ▶ Therefore the relevant tree is:



- ▶ The perfect equilibrium is: Player 1 chooses *b* and the strategy of 2 is to choose *f* if Nature chooses *U* and to choose *g* if Nature chooses *D*.
- ▶ The actual perfect equilibrium outcome is either :
 - ▶ $(-10, 40)$ or $(30, -30)$, equally likely. The expected payoffs are $(10, 5)$.

Example: 2-person game in tree form with moves from nature.

- ▶ The set of all Nash equilibrium points of this game
- ▶ We convert the game tree of Figure 10 into a game in strategic form.
This is a little tricky.
- ▶ Notice that Player 2 has four strategies: (e,g), (e,h), (f,g) and (f,h).
- ▶ Suppose that Player 1 chooses a. Let us compute the expected payoffs for each one of the four strategies:
 - (1) $(e,g) \rightarrow (3/10)*(10,0) + (7/10)*(0,30) = (3,21)$
 - (2) $(e,h) \rightarrow (3/10)*(10,0) + (7/10)*(40,10) = (31,7)$
 - (3) $(f,g) \rightarrow (3/10)*(0,20) + (7/10)*(0,30) = (0,27)$
 - (4) $(f,h) \rightarrow (3/10)*(0,20) + (7/10)*(40,10) = (28,13)$
- ▶ If 1 chooses b the expected payoffs are (No matter what 2 does):
- ▶ $(1/2)*(-10,40) + (1/2)*(30,-30) = (10,5)$



Example: 2-person game in tree form with moves from nature.

- We obtain the following table:

		2			
		(e, g) (e, h) (f, g) (f, h)			
1	a	3,21	*30,7	0,27 *	*28,13
	b	*10,5 *	10,5 *	*10,5 *	10,5 *

- There are two Nash equilibrium points:
 - (1) $(b, (e,g)) \rightarrow (10,5)$
 - (2) $(b, (f,g)) \rightarrow (10,5)^*$
- The second equilibrium is perfect, but not the first. In the first equilibrium Player 2 chooses e when Nature selects U. This is not a sensible action: choosing f is a better action for 2. Nevertheless, even if 2 chooses e and not f Player 1 is best off with b than with a, and the outcome is the same.

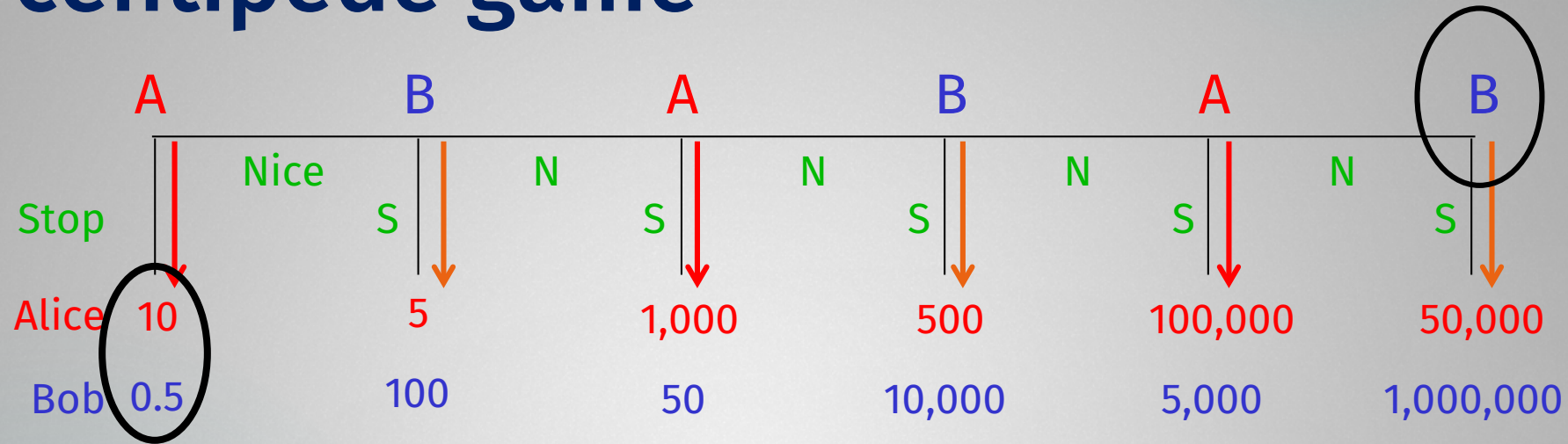
Evaluation of the perfectness concept

THE CENTIPEDE GAME

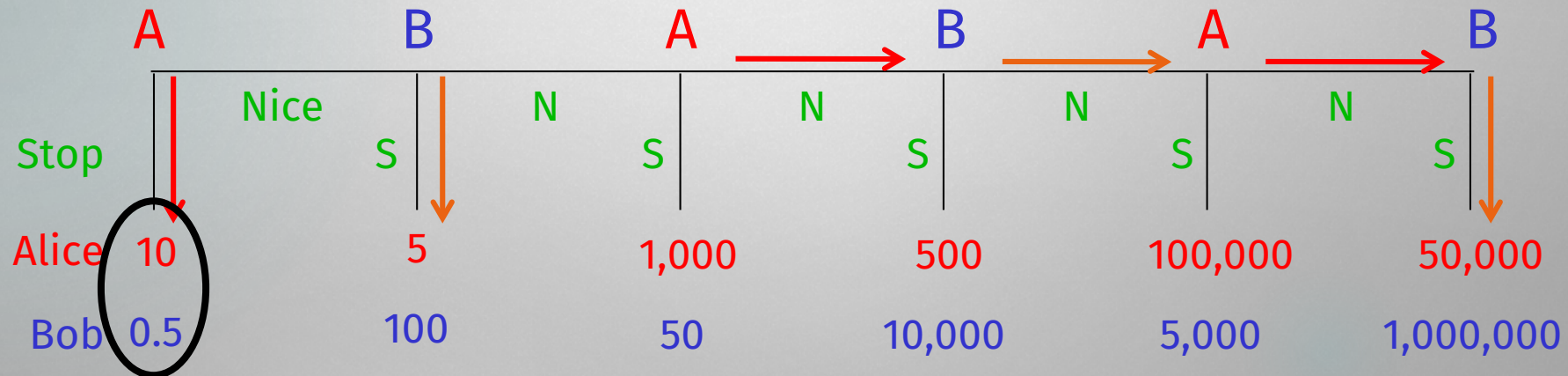
Criticism of the perfectness concept: The centipede game (Rosenthal 1981)

- ▶ Bob Rosenthal introduced this example to demonstrate the potential pitfalls of using the subgame perfect equilibrium concept to predict real human play *in some circumstances*.
- ▶ On the table there is a pot with \$10.5. Alice and Bob play the following game:
- ▶ Alice starts and she can either stop the game in which case they allocate the money in the pot where Alice gets \$10 and Bob gets \$0.5. But Alice can be nice to Bob and let him decide.
- ▶ In this case there is a miracle and the money in the pot increases ten times to \$105. If Bob decides to stop the game he takes \$100 leaving \$5 to Alice. But Bob can be nice to Alice and turn the decision to her.
- ▶ In this case the pot increases to \$1,050, etc. The game ends when in the pot we have at least \$1 million.

The centipede game



- ▶ Solving the game backwardly, the only perfect equilibrium outcome is (10, 0.5). Namely, Alice obtain \$10 and Bob \$0.5.
- ▶ Moreover, it can be shown that (10, 0.5) is the only equilibrium outcome.
- ▶ For example, another equilibrium is:



The centipede game: empirical results

- ▶ Several studies have demonstrated that the Nash equilibrium and likewise, the perfect equilibrium of the centipede game is rarely observed in real play.
- ▶ People that play the game in experiments regularly show partial cooperation, i.e. letting the game continue before eventually choosing to stop.
- ▶ However, it is rare for the players to cooperate through the whole game.
- ▶ Rosenthal (1981) suggested that if one has reason to believe her opponent will deviate from Nash Behavior, then it may be advantageous to not stop the game in the first round.
- ▶ Another explanation is that when the game is played for real, and the payoffs are understood in terms of money, the actual money gain might not indicate the real “utility” for the player: an example of this would be if people had altruistic preferences. We discuss this case in a previous forum.
- ▶ You had the chance to play this game in this week’s assigned experiments. What did you choose?

Simultaneous vs. sequential interactions

WHICH ONE IS BETTER?

Simultaneous vs. sequential interactions

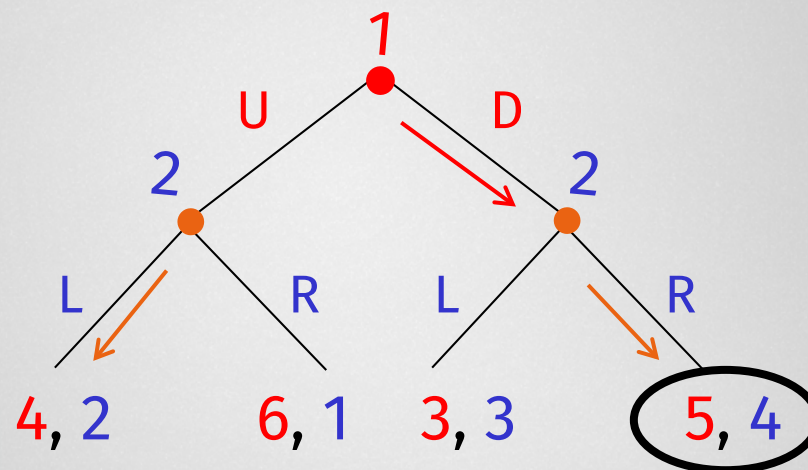
- ▶ There are cases where players that engaged in a simultaneous game (actions are taken simultaneously) are better off playing it sequentially.
- ▶ Example Consider the following two person (simultaneous) game in strategic form

		2	
		L	R
1	U	4, 2 *	6, 1
	D	3, 3	5, 4

- ▶ The strategy U of 1 strictly dominates the strategy D. Player 2 who knows that 1 will play U will choose L. The only sensible outcome is (4, 2).

Simultaneous vs. sequential interactions

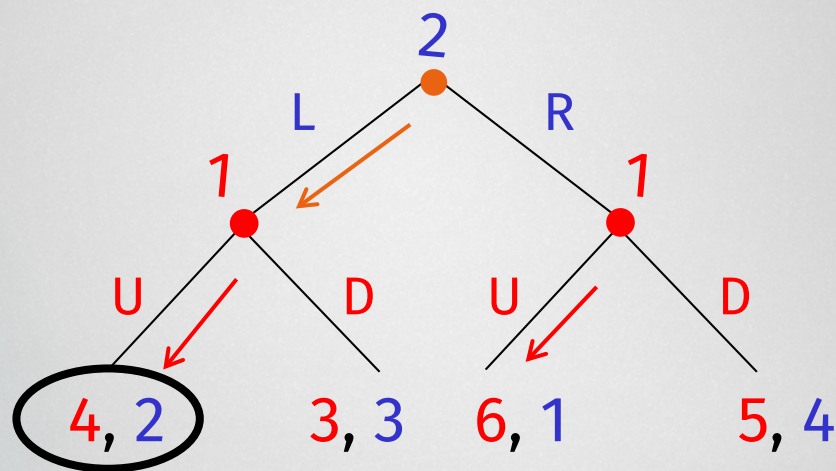
- Suppose next that the two players agree that 1 will move first and 2 will move second after observing the choice of 1. The new game is described in the tree below:



- Solving this game backwardly Player 1 chooses D and Player 2 chooses R and the perfect equilibrium outcome is now (5, 4). That is, both players are better off compared with the outcome (4, 2) of the simultaneous game.

Simultaneous vs. sequential interactions

- Suppose next that the two players agree that 2 moves first



- Here the perfect equilibrium outcome coincide with that of the simultaneous game.

Simultaneous vs. sequential interactions

- ▶ Consider the following game in strategic form

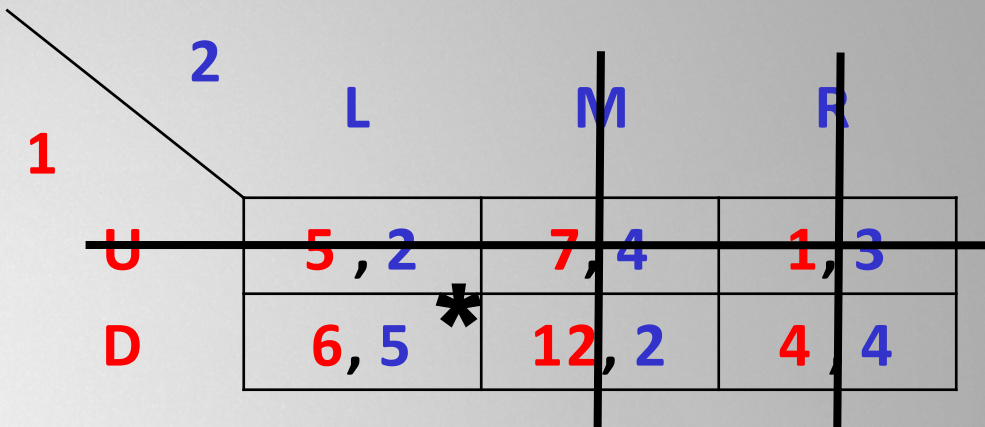
A strategic form game matrix for a simultaneous interaction between two players, 1 and 2. Player 1's strategies are U and D, and Player 2's strategies are L and R. The payoffs are listed in the cells, with the first number representing Player 1's payoff and the second number representing Player 2's payoff. An asterisk is placed above the (U, L) cell, indicating it is the only sensible outcome.

		2	
		L	R
1	U	5, 2	6, 1
	D	3, 3	4, 4

- ▶ The only sensible outcome is (5, 2). Verify that this outcome is also the perfect equilibrium outcome of the sequential game whether 1 moves first or 2 moves first.

Simultaneous vs. sequential interactions

- ▶ Consider the following game in strategic form
- ▶ The only equilibrium outcome is (6, 5). Suppose next that 1 moves first. The following is the tree of this game.
- ▶ The perfect equilibrium outcome is (7, 4). Player 1 improves his payoff (from 6 to 7) but the payoff of Player 2 decreases (from 5 to 4).



		2			
		L	M	R	
1	U	5, 2	7, 4	1, 3	
	D	6, 5 *	12, 2	4, 4	

